

NGHE-NHAN TRUONG (장예인)*

--- AI R&D Engineer ---

Date of birth: 13/09/2000
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About me

Computer Science Master's graduate specializing in **Deep Learning, Computer Vision**, and **Image Processing**. Experienced in developing robust solutions for image registration, multimodal fusion, object detection (2D, 6D), physical AI, and edge device optimization. As a dedicated researcher, the publication record includes **8 conference papers** (6 as first-author) and **2 first-author journal manuscripts in preparation.**

Education

❖ Sejong University (Seoul, South Korea) *Sep2023-Aug2025*

- Master of Science (M.S.) in Computer Science and Engineering
- **GPA: 4.33/4.5**
- Thesis: Transformer-based Facial Disguise Detection Using Multimodal Fusion of Visible and Thermal IR Images (Advisor: Prof. Seong Gon Kong)

❖ Ho Chi Minh City University of Technology and Education (HCMUTE) *2018-2022*

- Bachelor of Engineering (B.E.) in Electronics and Communications Engineering Technology
- Thesis: Research and Development of an **Interactive Vietnamese Sign Language Teaching System** for Hearing-Impaired Children.
- **Achievement: Best student** with the highest thesis score **(9.1/10)**

Honors and Awards

- **A fully funded Master program at Sejong University.** *Sep2023-Aug2025*
  *Awarded a full scholarship from Sejong University and a research grant from **IITP association** .*
- **Honor prize** for "Student of 5 merits" at HCMUTE *2022*
  "Molarity – Good Studying – Good Volunteer – Good Health – Good Integration"
- **Academic Excellence Scholarship** in 4 consecutive semesters at HCMUTE. *2020-2022*
  Given to top 3% students with highest GPA in each semester
- **Award** for Outstanding Student Achievement in Youth Month *2019*
- **Runner-up** in the National Science and Engineering Competition for High School Students. *2018*

Research Experience

❖ Imaging and Intelligent Systems Laboratory (Sejong University) *Sep2023-Sep2025*

Research Assistant – Lab Leader

Domain: **Deep Learning-based Video Surveillance, (Funded by IITP, South Korea)**

- **Project: Multimodal Facial Disguise Detection & Registration *(link)***
  - Physics-informed non-rigid registration pipeline for visible-thermal facial images.
  - Reduced affine/homography search space to 3 physical parameters (yaw, pitch, distance).
  - Noise-resilient rigid transformation utilizing Gaussian Field optimization.
  - Novel outlier rejection based on physics and rigid spatial locality consensus.
  - Proposed a facial disguise dataset and a multimodal thermal-visible fusion network via an iterative self-cross attention Transformer.
- **Project: Abnormal Behavior Detection *(link)***
  - YOLOv8-based detection and tracking pipeline with post-processing algorithm.
  - Sequential action recognition (e.g., loitering, abandonment, fall-down, violence) utilizing a Transformer architecture and ViTPose.

Research Experience

❖ HCMUTE Intelligent System Laboratory *(link)* *Jan2020–Jan2023*

Research Assistant

- **Project: Infrared and Thermal Fusion for Human Detection in Smoke Scenarios** ***(link)***
  - Precise point-to-point image alignment using the Light Heating Chessboard technique.
  - Real-time YOLOv4-based multimodal fusion deployed on NVIDIA Jetson Nano.
  - Created a diverse 3,000-image IR/Thermal dataset simulating fire and human rescue scenarios.
- **Project: Vietnamese Hand-Sign Language Teaching System** ***(link)***
  - Static and dynamic gesture tracking pipeline leveraging YOLOv5, GRU, and MediaPipe.
  - Designed, developed a fully custom 3D-printed electro-mechanical prosthetic arm for synchronized gesture execution.

Industry Experience

❖ Yuil Robotics *Sep2025–Present*

AI Research Engineer – Professional Manager (PM)

- **Technical Leadership & Architecture**:
  - Defined the **Vision team's** technical roadmap by translating **SOTA research** into deployable AI pipelines.
  - Architected **head-to-PC** dataflow and standardized the robotics software stack using a **Zenoh** publish-subscribe mechanism.
- **AI Research & Development**:
  - **6D Pose Estimation:** Optimized Foundation Pose to achieve at 20 FPS on Jetson AGX Orin.
  - **Embodied AI (VLA):** Fine-tuned the $\pi_{0.5}$ model for manipulation on dual-arm setups.
  - **Sim-to-Real Transfer:** Domain Randomization, Co-training, Cosmos Augment, and GapOnet.
  - **Hand-Eye Calibration:** Built an automatic online calibration pipeline for dual-arm robots.

❖ GaraSTEM Educational Technology Joint Stock Company *Jan2023–Jul2023*

R&D Intern, Entry-Level Engineer

- **Responsibility**:
  - Developed deep learning models for an automated commercial attendance system.
  - Built an API to integrate visual recognition models with web services.

Teaching Experience

❖ Teaching Assistant *Sep2023-Sep2025*

Sejong University

Course: Advanced FPGA with Zybo-Z7, Embedded Computing with EFM8BB1.

- Designed course curricula and structured lecture materials.
- Developed practical demo codes to reinforce theoretical concepts.
- Assessed student learning through assignments and projects.

❖ STEM Instructor *Jan2023–Jul2023*

GaraSTEM Educational Technology Joint Stock Company

- Instructed high school students in fundamental STEM concepts and practical applications
- Organized STEM competitions and provided technical mentorship to student teams.

Activities

- **Singer for Internation conference**: Peace – Aspiration of Ancestor 2024
- **Certificate** of online 60km marathon contest *2022*
- **The 1st prize** in the duet of the Singing Talent Contest of Ho Chi Minh City *2021*
- **Green Summer Campaign** organized by Ho Chi Minh Communist Youth Union *2018 & 2020*
- **Spring Volunteer Campaign** organized by Ho Chi Minh Communist Youth Union *2018 & 2020*
- **President** of High school Science and Technology Club *2015-2018*

Skills

- **Foreign Languages:** **English** (Ielts 6.0), Korean (intermediate)
- **Programming Languages:** Python, C/C++, MATLAB
- **Frameworks:** TensorFlow, PyTorch, Paddle, Fast API, PyQT designer, Docker, Issac-Sim, Issac-Lab.

References

Prof. **Seong Gon Kong**
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Publications

(† ***Equal contribution / Co-first authors)***

❖ **Peer-Reviewed Publications**

1. **N. -N. Truong**, T.-D. Do, L.-A. Tran, T.-D. Nguyen, and M.-H. Le, "Efficient Infrared and Thermal Imaging Fusion Approach for Real-time Human Detection in Heavy Smoke Scenarios," 2023 International Conference on System Science and Engineering (ICSSE), Ho Chi Minh City, Vietnam, 2023. *(link)*
2. **N. -N. Truong**, T.-D. Do, T.-S. Le Nguyen, M.-T. Duong, T.-H. Nguyen, and M.-H. Le, "A Vision-based Hand-sign Language Teaching System using Deep Neural Network: Methodology and Experiments," 2022 International Workshop on Intelligent Systems (IWIS), Ulsan, Korea, 2022, pp. 1-6. *(link)*
3. D. N. -L. Vy†, **N. -N. Truong**† , V. D. Nguyen, T. -D. Dao and M. -H. Le, "Hybrid Agentic Architecture for Vietnamese Sign Language: Orchestrating RAG and Syntactic Restructuring," 2026 8th International Conference on Green Technology and Sustainable Development (GTSD), Ho Chi Minh City, Vietnam, 2026 *(link)*
4. V. D. Nguyen†, **N. -N. Truong**†, T. -D. Dao, D. N. -L. Vy and M. -H. Le, "TKR-HMR: Tokenized Kinematic Representation for Human Mesh Recovery from Video," 2026 8th International Conference on Green Technology and Sustainable Development (GTSD), Ho Chi Minh City, Vietnam, 2026 *(link)*
5. T. -D. Dao†, **N. -N. Truong**†, D. N. -L. Vy, V. D. Nguyen and M. -H. Le, "From Multi-View RGB-D to Single-View: Fusion and Distillation for Efficient Sign Language Recognition," 2026 8th International Conference on Green Technology and Sustainable Development (GTSD), Ho Chi Minh City, Vietnam, 2026 *(link)*
6. T. -D. Do, L. -A. Tran, T. -D. Nguyen, **N. -N. Truong**, D.-C. Park, and M.-H. Le, "POCS-based Image Compression: An Empirical Examination," 2024 7th International Conference on Green Technology and Sustainable Development (GTSD), Ho Chi Minh City, Vietnam, 2024. *(link)*
7. M. -T. Duong, T. -D. Phan, **N. -N. Truong,** M.-C. Le, T.-D. Do, V.-B. Nguyen, and M.-H. Le, "An Image Enhancement Method for Autonomous Vehicles Driving in Poor Visibility Circumstances," Computational Intelligence Methods for Green Technology and Sustainable Development, Lecture Notes in Networks and Systems, vol. 567, Springer, Cham, 2023. *(link)*

Publications

(† *Equal contribution / Co-first authors)*

❖ **Manuscripts Submitted & In Preparation**

1. **N. -N. Truong**, and S. -G. Kong, “Physics-informed Multimodal Images Registration for Video Surveillance”, (Manuscript in preparation, target journal: **Information Fusion**)
2. **N. -N. Truong**, and S. -G. Kong, “Transformer-based Facial Disguise Detection Using Multimodal Fusion of Visible and Thermal IR Images”, (Manuscript in preparation, target journal: **Information Fusion**)
3. V. D. Nguyen†, **N. -N. Truong**†, T.-D. Do, and M. -H. Le, "QANet-Lite — Quad-Axial Lightweight Attention Network for Keyword Spotting," 2026 1st n Electrical, Electronics and Intelligent Systems (ICEEIS), Ho Chi Minh City, Vietnam, 2026 (**Accepted**)